Shreya Aggarwal
CU24250082
Roll No: 58
Section: C

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Start coding or generate with AI.
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```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image_path = "/content/Panda.png"

img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

plt.figure(figsize=(6, 5))
plt.imshow(img, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

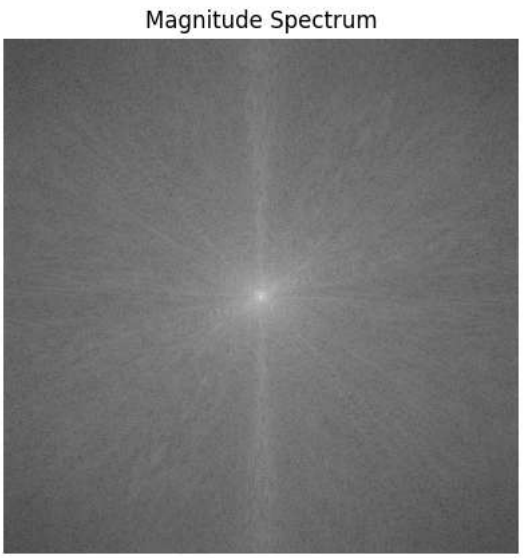



```
dft = np.fft.fft2(img)
```

```
dft_shift = np.fft.fftshift(dft)
```

```
magnitude_spectrum = 20 * np.log(np.abs(dft_shift) + 1)

plt.figure(figsize=(6, 5))
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis("off")
plt.show()
```




```
rows, cols = img.shape
crow, ccol = rows // 2, cols // 2

mask_low = np.zeros((rows, cols), np.uint8)

radius = 30
cv2.circle(mask_low, (ccol, crow), radius, 1, -1)

low_pass = dft_shift * mask_low
```

```
low_pass_shift = np.fft.ifftshift(low_pass)

low_pass_img = np.fft.ifft2(low_pass_shift)
low_pass_img = np.abs(low_pass_img)

plt.figure(figsize=(6, 5))
plt.imshow(low_pass_img, cmap='gray')
plt.title("Low-Pass Filtered Image")
plt.axis("off")
plt.show()
```




```
mask_high = np.ones((rows, cols), np.uint8)

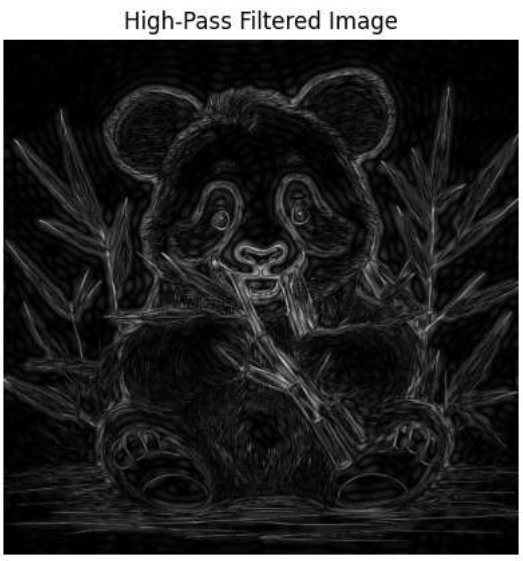
cv2.circle(mask_high, (ccol, crow), radius, 0, -1)

high_pass = dft_shift * mask_high
```

```
high_pass_shift = np.fft.ifftshift(high_pass)

high_pass_img = np.fft.ifft2(high_pass_shift)
high_pass_img = np.abs(high_pass_img)

plt.figure(figsize=(6, 5))
plt.imshow(high_pass_img, cmap='gray')
plt.title("High-Pass Filtered Image")
plt.axis("off")
plt.show()
```




```
plt.figure(figsize=(12, 8))

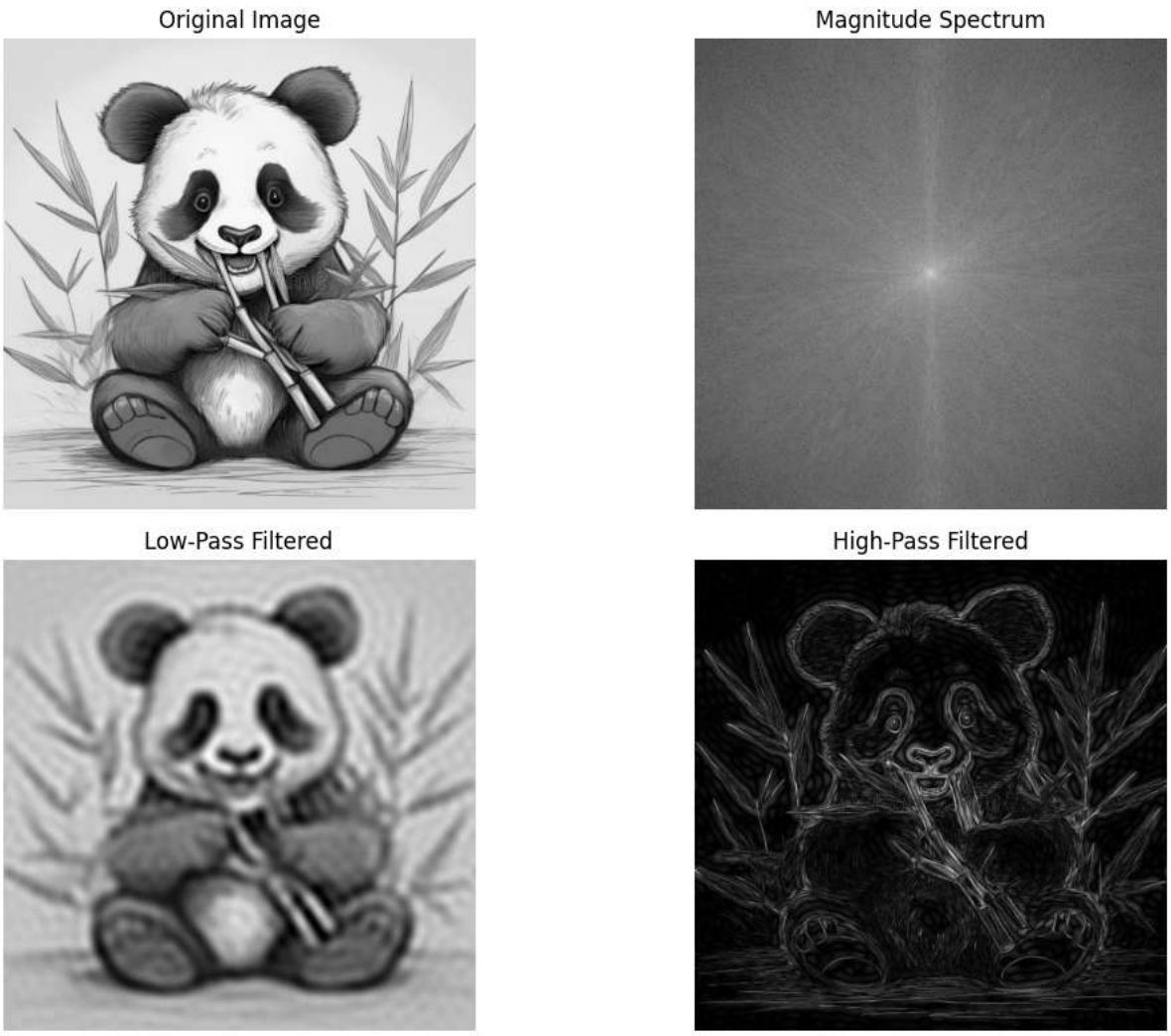
plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Magnitude Spectrum")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(low_pass_img, cmap='gray')
plt.title("Low-Pass Filtered")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(high_pass_img, cmap='gray')
plt.title("High-Pass Filtered")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Observations: All the image information is present in the original image. Magnitude Spectrum indicates the frequencies of the image. Low Pass filter gives smooth image by eliminating high frequencies from the image. High Pass filter highlights the edges and finer details in an image.

Conclusion: Fourier transform has been applied successfully on the image, and Low Pass and High Pass Frequency filters have been done successfully.

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